



**SECV2113**  
**HUMAN COMPUTER INTERACTION**  
**SECTION 1**

**ASSIGNMENT 1**

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## Purpose of the Technology



The global goals which our group presenting is the Quality Education Goals. This global goal is to ensure inclusive and equitable quality education and promote lifelong learning opportunities for everyone. Education liberates the intellect, unlocks the imagination and is fundamental for self-respect. It is the key to prosperity and opens a world of opportunities, making it possible for each of us to contribute to a progressive, healthy society. Learning benefits every human being and should be available to all.

Resources: <https://www.globalgoals.org/goals/>

There are ten targets which can help us to make sure that we meet the Global Goals such as free primary and secondary education, universal literacy and numeracy, increase the supply of qualified teachers and others. In this, we are going to introduce a technology that used by us daily which is e-learning, an application who used for education in most of university in Malaysia.



The primary objective of the e-learning application is to enable online education and training. This platform offers users the opportunity to engage with educational resources, enroll in every course, providing a platform to communicate with educators and fellow learners, undertake assessments, submit assignments to instructors and monitor their learning journey through digital channels. The e-learning application is designed to make learning more accessible, flexible and convenient which can allow individuals to participate in education regardless of their location and time.

Other than that, this application is also personalizing the learning experience for every user. Through the system, the application can classify the learner by their courses and able them to access to their correct course material and classes. Lastly, this application is also cost-effectiveness compared to the traditional classroom-based education as it eliminates expenses of physical materials and facilities maintenance.

In e-learning systems, designers are using persuasive design to boost user engagement and learning. They're focusing on three key factors: motivation, ability, and triggers. By understanding how these factors influence behaviour, they're creating features that make learning easier and more motivating.

**Feature 1:** To send prompt notifications to students.

**Target behaviour:** Click on the notification to check the details and submit works before the deadline set by the lecturer.

**Ability:**

Prompt notifications can increase students' ability to complete their schoolwork by providing timely reminders and easy access to resources. For example, if a student clicks on the prompt notification, the user interface will take users to the relevant page which showing the work instructions and attachments provided by lecturers. Moreover, timely notifications can assist students in better time management by prompting them to begin their homework earlier. This can enhance their chances of finishing it well before the deadline. In a nutshell, push notification to students regarding their homework assignments can boost their task completion by offering timely reminders, access to resources and support. This significantly contributes to improving their overall academic performance.

**Motivation:**

These notifications keep the students motivated to stay on track with their schoolwork when they receive these prompt notifications. The reminder acts as a gentle push to encourage them to take action and finish their schoolwork promptly. Furthermore, notifications could serve as motivation for students aiming to achieve good grades, as most university assignments significantly impact their final course grades. When students get timely notifications from the educational app regarding upcoming homework deadline, it will boost their drive to finish tasks without delay. This reminder helps students stay focused on their study objective and prompts them to take the required actions, such as scheduling dedicated study time and finishing their homework before the due date. In essence, these prompt notifications assist students in staying motivated and responsible for their academic tasks.

**Trigger:**

The push notifications serve as a reminder by prompting students to remember their homework tasks and trigger them to take on them. When students receive a prompt notification about their homework, it acts as a encourage that activates their awareness of the task at hand which is already close to the deadline. In this way, the notification work as triggers that activate students' memory and persuade them to complete their schoolwork. Ultimately, they serve as timely reminder that help students stay organized and keep themselves on track with their academic obligations.

**Feature 2:** Add submission button.

**Target behaviour:** Click “Add Submission” button to submit files and assignment.

**Ability:**

The user’s ability to click the "Add submission" button is influenced by the interface design. The button is prominently displayed on the dashboard or assignment page, labelled clearly, and easily distinguishable from other elements, it enhances the user’s ability to locate and click it. The design is intuitive as it requires minimal effort on the user’s part to understand how to add an assignment. The button is labelled with clear and concise language, “Add submission” and is visually distinct, users are more likely to recognize and interact with it.

After a click, students are directed to a subsequent page where they are prompted to upload their files and assignments, the details are clearly stated at the same page. Upon successful file upload, students are required to execute the final step of submission by clicking the "Save changes" button, or alternatively, they may choose to exit by selecting the "Cancel" button.

The steps are designed to be user-friendly that ensuring users have a high ability to achieve the target behaviour.

**Motivation:**

The user's motivation to click the "add submission" button could stem from various factors. Students may be motivated to submit assignments because they are genuinely interested in the subject matter or because they want to demonstrate their understanding and skills. Besides, students are motivated by the desire to make progress towards completing their task. Clicking the "Add Submission" button signifies the first step in this process, motivating the students to initiate the upload of their files and assignments. Furthermore, the fear of negative consequences associated with not completing the submission or missing deadlines may be the motivation of students to achieve the target behaviour. Clicking the "Add Submission" button serves as a proactive measure to avoid such consequences.

**Trigger:**

In the e-learning system, triggers could be notifications or reminders prompting the user to submit an assignment. To maximise the potential for action, these triggers should be appropriately timed and contextually relevant. For instance, receiving a notification that an assignment deadline is approaching can serve as a trigger. Time-sensitive reminders, such as impending deadlines or expiration of submission windows, can serve as triggers that prompt students to click the "Add Submission" button. The urgency created by these time constraints motivates students to take immediate action. The "Add Submission" button, prominently displayed and clearly labelled, acts as a definitive trigger prompting users to commence the submission process. Its conspicuous placement and informative label encourage students to initiate the action of uploading their files and assignments without hesitation.

However, despite these successful implementation, some e-learning system features have missed the mark in achieving the desired user behaviour. These features may not effectively tap into motivation, simplify tasks for user ability and prompt action through triggers, resulting in not ideal outcome.

**Feature:** The grades section.

**Target behaviour:** Click to see or check students' grades for each assessment.

**Ability:**

Simplifying the grade section can improve user ability by making it easier for lecturers to navigate and interpret their students' grades. For students, instead of sorting through multiple assignments or assessments to check their grades, they can quickly access their final grade for each course, saving time and effort. As now, lecturers are interpreting students' grades through another website or by sending the name list with grades for students, so I could say that this is a better way that could improve their ability to use this feature. Furthermore, providing clear instructions or tips can further enhance user ability by providing guidance on how to interpret the final grade and its significance within the context of the course.

**Motivation:**

This section lets students check their marks and grades, but it seems to be too packed and overloaded. Simplifying the grade section to display only the final grade for each course of each semester rather than each and every assignment and exercise would be better. By displaying only the final grades, it can enhance motivation by reducing cognitive load, space and streamlining the information presented to users. By knowing their overall performance in a course briefly can provide students a clearer understanding of their progress and performance, which may motivate them to strive for improvement or maintain their current level of achievement.

**Trigger:**

Presenting the final grade prominently in the grade section can serve as a trigger to prompt students to reflect on their performance and take action if necessary. Seeing their final grade may prompt students to seek additional assistance, review course materials, or adjust their study strategies. Additionally, integrating notifications or reminders related to grade updates, such as when final grades are published or when there are significant changes in performance, can further serve as triggers to encourage student engagement with the grade section.

**CHAU YING JIA** Message

**Courses I am taking**

| Course name                                                                                    | Grade |
|------------------------------------------------------------------------------------------------|-------|
| ULRS1182-10 PENGHAYATAN ETIKA DAN PERADABAN (APPRECIATION OF ETHICS AND CIVILIZATIONS)         | -     |
| SECV2113-01 INTERAKSI MANUSIA KOMPUTER (HUMAN COMPUTER INTERACTION)                            | -     |
| SECI1143-03 KEBARANGKALIAN STATISTIK & ANALISIS DATA (PROBABILITY & STATISTICAL DATA ANALYSIS) | -     |
| SECR1033-03 ORGANISASI & SENIBINA KOMPUTER (COMPUTER ORGANIZATION AND ARCHITECTURE)            | -     |
| SECR1033-02 ORGANISASI & SENIBINA KOMPUTER (COMPUTER ORGANIZATION AND ARCHITECTURE)            | -     |
| SECP2613-01 ANALISIS DAN REKABENTUK SISTEM (WBL) (SYSTEM ANALYSIS AND DESIGN (WBL))            | -     |
| SECJ1023-02 TEKNIK PENGATURCARAAN II (PROGRAMING TECHNIQUE II)                                 | 4.00  |
| PEMBELAJARAN AI UNTUK RAKYAT AI2024-01                                                         | -     |

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**Grades**

| Grade item                                                                                                                                                              | Calculated weight | Grade | Range | Percentage | Feedback | Contribution to course total |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------|-------|------------|----------|------------------------------|
| ▼ ULRS1182-10 PENGHAYATAN ETIKA DAN PERADABAN (APPRECIATION OF ETHICS AND CIVILIZATIONS)                                                                                |                   |       |       |            |          |                              |
| ASSIGNMENT<br>UPLOAD BORANG PERINCIAN MARKAH DI SINI. KEKALKAN DALAM WORD. RENAME FILE NAMA SENDIRI HURUF BESAR. TERIMA KASIH.                                          | 0.00 %<br>(Empty) | -     | 0-100 | -          |          | 0.00 %                       |
| ASSIGNMENT<br>UPLOAD TUGASAN ARTIKEL AKHBAR DI SINI DALAM PDF. RENAME FILE NAMA KUMPULAN CONTOHNYA KUMPULAN 8. HANYA KETUA,ATAU WAKIL SAHAJA YANG UPLOAD. TERIMA KASIH. | 0.00 %<br>(Empty) | -     | 0-100 | -          |          | 0.00 %                       |
| ASSIGNMENT<br>UPLOAD JAWAPAN UJIAN DI SINI DALAM BENTUK PDF. RENAME FILE NAMA SENDIRI HURUF BESAR. TERIMA KASIH.                                                        | 0.00 %<br>(Empty) | -     | 0-100 | -          |          | 0.00 %                       |
| AGGREGATION<br>Course total                                                                                                                                             | -                 | -     | 0-100 | -          |          | -                            |

This picture shows the grades section in our e-learning website. It shows the final grades for each course, but lecturers must update all the marks of assessment to get this score. Based on the second picture, it shows the marks for each assessment. If the structure could be simplified would be better and user-friendly.

## Conclusion

To sum up, e-learning system have significantly contributed to democratizing education by remove restriction on location and make learning opportunities available to individuals worldwide. The analysis of e-learning system features highlights both successes and shortcomings in achieving target behaviours that influenced by the factors of ability, motivation and triggers. In order to improve e-learning system and enable meaningful learning experience, it will be crucial to incorporate a greater understanding of user motivations, abilities and design or enhance a better function as a trigger to execute the target behaviour.

## References

Fogg, Brian J. "A behavior model for persuasive design." *Proceeding of the 4<sup>th</sup> international Conference on Persuasive Technology*. ACM, 2009.

Project Everyone. "The 4<sup>th</sup> Global Goals, Equality Education" Retrieved from:  
<https://www.globalgoals.org/goals/4-quality-education/>

Elearning. (n.d.). Retrieved from: <https://elearning.utm.my/>